

Design Spec

Cell Design Specification

Virtual Battery Factory · case t7_r1 · 2026-08-25

Cell Design Specification — HEV Fast-Charge Battery (Case t7_r1)

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1. Basic Specification

Item	Value	Source
Electrochemical system	NMC811 / graphite (pouch, Chen2020 baseline parameterization)	entry 0 meta.base; anchor table
Nominal capacity	6.0313 Ah	params_v11.json (mechanically = measured 1C capacity, rule per propose_r2b)
Measured 1C capacity (DFN)	6.032209 Ah	cell/r5_v11_1c_dfn.json:capacity_ah
Voltage window	2.5 – 4.2 V	Chen2020 parameter set (Lower/Upper voltage cut-off)
Nominal (midpoint) voltage	3.7237 V	cell/r5_v11_energy.json:midpoint_voltage_v
Cell dimensions (electrode stack)	height 65 mm × width 1580 mm × thickness 230.6 μm	parameter set (Electrode height/width); thickness = calc-energy sum
Shell / enclosure thickness	Not provided	no parameter in set
Electrolyte formulation	EC/EMC + LiPF6 base; transport overrides $\sigma = 2.5$ S/m, $t_{\text{H}^+} = 0.55$, $D_{\text{e}} = 6 \times 10^{-14}$ m ² /s	params_v11.json (design); base = Chen2020
Electrolyte additive candidates	None (Stage 2 not entered; start_stage = 3, no new materials in task)	entry 0 meta
Cation transference number	0.55	params_v11.json
Cooling design	$h = 50$ W/(m ² ·K) on cell cooling surface 0.00531 m ² → $hA = 0.2655$ W/K	params_v11.json; bridge (area = "Cell cooling surface area [m2]", Chen2020/injected default)

2. Electrode and Separator

Layer	Thickness	Porosity	Particle radius	AM volume fraction	Density (set)	Current collector
Positive (NMC811)	75.6 μm	0.335	3.5 μm (design)	0.665	3262 kg/m ³	Al 16 μm (2700 kg/m ³)
Separator	12 μm	0.47	—	—	397 kg/m ³	—
Negative (graphite)	115 μm (design)	0.36 (design)	1.5 μm (design)	0.75	1657 kg/m ³	Cu 12 μm (8960 kg/m ³)

- All values dumped from the Chen2020 parameter set with params_v11 overrides applied (mechanical dump, 2026-08-25).

- N/P ratio = negative theoretical areal capacity ÷ cell measured areal capacity (positive usable capacity):

- Negative theoretical areal capacity = $115 \mu\text{m} \times (1 - 0.36) \times 33133 \text{ mol/m}^3 \times F/3600 = 65.36 \text{ Ah/m}^2$

- Cell measured areal capacity = $6.032209 \text{ Ah} \div 0.1027 \text{ m}^2 = 58.74 \text{ Ah/m}^2$

- Practical N/P = 1.113 (full-range theoretical N/P = $65.36/58.74 = 1.113$, positive full-range 85.03 Ah/m^2 — NMC811 utilises ≈69 % of its theoretical range in the 2.5–4.2 V window; both values reproduced in calc.xlsx)

- Maximum concentrations: positive 63104 mol/m³, negative 33133 mol/m³ (parameter set).

3. Process Design Parameters

Parameter	Formula	Value	Notes
Positive areal density	$l \times (1 - \epsilon) \times \rho$	$75.6 \text{e-}6 \times 0.665 \times 3262 = 164.0 \text{ g/m}^2$	parameter set
Negative areal density	$l \times (1 - \epsilon) \times \rho$	$115 \text{e-}6 \times 0.64 \times 1657 = 122.0 \text{ g/m}^2$	parameter set
Positive compaction density	$\rho \times (1 - \epsilon) / 1000$	$3262 \times 0.665 / 1000 = 2.17 \text{ g/cm}^3$	set values
Negative compaction density	$\rho \times (1 - \epsilon) / 1000$	$1657 \times 0.64 / 1000 = 1.06 \text{ g/cm}^3$	set values
Electrolyte fill amount	pore volume × ρ_{elec} × fill factor	$7.432 \text{ cm}^3 \times 1.2 \text{ g/cm}^3 \times 1.0 = 8.92 \text{ g}$	$\rho_{\text{elec}} = 1.2 \text{ g/cm}^3$ literature value; fill factor 1.0 assumption
Formation recommendation	0.1C CC charge to 4.2 V, 25 °C, 2 cycles	—	design recommended value; production-line value requires tuning

4. Mass Breakdown (formula caliber = calc-energy contract, electrolyte excluded)

Layer	Mass (g)	Formula / source
Positive electrode coating	16.842	layer_kg_m2 0.163994 × 0.1027 m ² (cell/r5_v11_energy.json)

Layer	Mass (g)	Formula / source
Negative electrode coating	12.525	layer_kg_m2 0.121955 × 0.1027 m²
Positive current collector (Al)	4.437	0.0432 × 0.1027
Negative current collector (Cu)	11.042	0.10752 × 0.1027
Separator	0.259	0.00252492 × 0.1027
Total (contract, electrolyte excluded)	45.105	cell/r5_v11_energy.json:mass_kg
Electrolyte (additive, estimate)	8.92	pore-volume formula above, not in contract mass

5. Performance Verification (vs entry 0 criteria)

Metric	Threshold	Value	Verdict	Source
Energy density	≥ 327.18 Wh/kg	483.247 Wh/kg	✓ PASS	cell/r5_v11_energy.json:energy_density_wh_kg
4C fast charge plating	plated = false	anode surface min +0.0476 V → not plated	✓ PASS	cell/r5_v11_4c45.json:anode_potential_v
4C max temperature	not adjudicated (absent from task)	330.130 K (56.98 °C) at 45 °C ambient	informational	cell/r5_v11_4c45.json:T_max_K
SEI after 100 cyc @45 °C	≤ 550 nm	465.473 nm	✓ PASS	cell/r5_v11_aging45.json:sei_thickness_nm_end
Nail penetration (10 W) TR	triggered = false	triggered = false, T_max 336.320 K, dT/dt max 0.245 K/s	✓ PASS	cell/r5_v11_nail.json
1C capacity	— (not adjudicated)	6.032209 Ah	informational	cell/r5_v11_1c_dfn.json:capacity_ah

6. Design Notes (what changed vs baseline and why)

Round	Change	Rationale (evaluate log)
R1	Baseline Chen2020	ED 400.75 ✓, SEI 476.1 ✓, but 4C plating (anode min −0.1918 V) and nail TR (triggered 322 s) FAIL
R2	Transport set V1–V3 (σ/t■/D_e, h)	plating persists (−0.134 V best); V4 ceiling probe ED 464 Wh/kg → ED non-binding, no Stage-2 escalation
R3	Isolations: porosity dominant (−0.134 → −0.0419 V); h=200 worsens plating	V5 combination (+0.0351 V, porosity 0.32, 1.5 μm graphite, D_e 5e-10, h 50) first full pass
R4	Margin widening: V8 σ2.5/t■0.55/D_e6e-10/poro0.36 (+0.0494 V)	nail cooling floor mapped: h=40 passes (T_max 348.7 K), h=30 triggers at 666 s (r5_v8_nail_h30)
R5	V11-final = V8 + h=50 (nominal mechanically 6.0313)	DFN-precision verification: all four criteria pass (this sheet, §5)